CycleTwin

Synthetic cycle forecasting for clinician decision support

Digital Twin Challenge 2026 · Kushagra Gupta · VIT

Synthetic engineering prototype · no clinical validation

Not for diagnosis, contraception or conception planning.

The forecasting question

Can a system flag a likely delayed or absent ovulation early in a cycle, while showing uncertainty and limits?

- Target: synthetic anovulation or ovulation after cycle day 21
- Forecast points: cycle days 10, 14 and 18
- Decision-support framing; no diagnostic or reproductive-planning use

PMOS is a multi-system condition

Polyendocrine metabolic ovarian syndrome (PMOS; PCOS/PCOD)

The project frames the problem across reproductive and metabolic signals. The synthetic build currently evaluates only a reproductive-axis proxy.

Terminology follows Teede et al., The Lancet, 2026. The simulated phenotype labels are not diagnostic criteria.

Why a digital-twin framing

Planned design

- Reproductive follicle and ovulation dynamics
- Metabolic glucose dynamics
- Shared insulin-resistance latent with reverse cycle coupling

Current build

- Stochastic synthetic cycle generator
- Fixed risk score and temperature coverline heuristic
- Simplified coupled twin shown in dashboard; not calibrated

Data fusion and sensor tiers

Tier 0

Period logs

›

Tier 1

+ resting HR*

›

Tier 2

+ nightly temp

›

Tier 3

+ CGM windows

›

Tier 4

+ EHR labs

Implemented evaluator: neutral period-only score; tier 1 uses resting-heart-rate offsets; tier 2 adds nightly temperature.

* Other wearable streams are generated but not fused. CGM display is synthetic only; E2 and EHR tiers are not evaluated.

Synthetic data and leakage control

- 1,200 synthetic patients × 6 cycles per generator seed
- Three fresh seeds: 97, 101, 103
- Patient-level split: 70% train, 15% validation, 15% held-out test
- No personal exports, credentialed data or external health records

The generator carries latent traits and noisy observations; the evaluator separates patients across partitions.

Forecast model

Risk score

- Tier 0: neutral baseline
- Tier 1: resting-heart-rate offset
- Tier 2: temperature rise relative to an early-cycle baseline

Ovulation-day estimate

- Temperature coverline heuristic
- Fallback: forecast day + 10
- Dashboard twin is separate and uncalibrated; no EKF

Conformal coverage is not reported. All settings are simulator assumptions.

Safety and abstention

Dashboard safety rules

- Review for thyroid/prolactin flags
- Abstain for contraception and pregnancy-related states
- Abstain for age and sensor completeness conditions
- Dashboard shows a synthetic risk-coverage diagnostic

Current status

G2 dashboard implements explicit abstention guards. Safety performance and conformal coverage are not evaluated.

Synthetic held-out results

Tier 2 · forecast day 14 · held-out patients

Seed 97: AUROC 0.637 · AUPRC 0.598 · Brier 0.306 · event rate 0.506

Seed 101: AUROC 0.638 · AUPRC 0.604 · Brier 0.316 · event rate 0.523

Seed 103: AUROC 0.663 · AUPRC 0.615 · Brier 0.309 · event rate 0.511

Patient-cluster bootstrap 95% intervals for AUROC and AUPRC are in results.json. Synthetic generator performance only.

No comparison with real-world published accuracy is made.

Illustrative counterfactuals

Planned scenario inputs

- Regular 7–8 hour sleep
- Daily 30-minute walk
- Lower-glycaemic-index Indian food swap
- Lower evening stress
- Illustrative 5% weight change

Current status

The dashboard shows hand-set risk ranges. No twin-based or generator counterfactuals are implemented. Effects are assumptions, not treatment advice.

Planned label: illustrative assumption, not a treatment recommendation.

Robustness and failure modes

- Current robustness sweeps are not implemented
- No wearable dropout, fever, travel or shift-work stress tests
- No simulator-shift experiment
- Known limitation: generator labels and signals share the same simplified assumptions

The synthetic benchmark cannot establish performance under real sensor noise or clinical variation.

Dashboard and Apple Health input

- Streaming XML parser for a limited HealthKit export subset
- Risk-ranked synthetic list, timeline, abstention banner and alert log
- Synthetic CGM panel, marginal conformal range and illustrative what-if controls
- CSV adapter supports date, temp, rhr, hrv, steps, period_start

The synthetic export is a schema-compatibility smoke test, n=1; it clinically validates nothing.

Related work

- Goodale 2019 · Luo 2020 · Wang 2025: wearable signals, differing prediction targets
- Awoniran 2026 preprint · vaginal temperature; Shamanna 2025 · single-patient twin case
- Lin 2023 · cycle glucose; Teede 2023 guideline and 2026 PMOS name consensus
- Gibbs & Candès 2021 · adaptive conformal; Collins 2024 · TRIPOD+AI
- mcPHASES is a restricted-access roadmap source; no data are used

These works inform terminology, design and reporting. They do not validate this synthetic simulator.

Limits and next steps

- Calibrate and connect the coupled twin to the E1 forecast
- Evaluate abstention and marginal conformal coverage on untouched data
- Add CGM/EHR prediction tiers, counterfactual simulation and robustness tests
- Seek clinician co-design and real-data validation before any clinical use

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